REPORT

Course Code: CSE 4408

Course Name: System Analysis and Design

Lab Number: 11

Project Title: BashayJabo

Group: Les Misérables

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Key System Outputs

1. Ride Confirmation Page

- **Intended User:** Ride creator/joiner
- **Primary Purpose:** Confirm that a ride has been successfully created or joined, preventing duplicate actions and ensuring user confidence.
- **Delivery Method:** Mobile app screen display + optional push notification.
- **Design Justification:**
  - Serve the intended purpose: Provides immediate confirmation of core action.
  - Fit the user: Simple, real-time display tailored for students.
  - Deliver the appropriate quantity: Shows only essential details (time, location, seats, participants).
  - Ensure output is where it is needed: Appears directly in the app post-action.
  - Provide output on time: Instant confirmation right after submission.
  - Choose the right output method: Screen display ensures accessibility, with optional push for reminders.

2. Ride Listings View

- **Intended User:** Ride seekers
- **Primary Purpose:** Display all current ride offers with pickup, destination, times, and available seats, enabling informed ride selection.
- **Delivery Method:** In-app screen (card view) with filtering/sorting capabilities.
- **Design Justification:**
  - Serve the intended purpose: Centralized browsing solves the current problem of scattered WhatsApp/Facebook posts.
  - Fit the user: Student-friendly UI with filters (gender, department, time).
  - Deliver the appropriate quantity: Presents only active rides, avoiding overload.
  - Ensure output is where it is needed: Accessible within the app dashboard.
  - Provide output on time: Rides update dynamically as new offers are added.
  - Choose the right output method: Interactive listing view with filters is intuitive for mobile.

3. Payment Receipt / Transaction Log

- **Intended User:** Students making/tracking payments
- **Primary Purpose:** Record completed transactions, showing fare split, payment method (Bkash, Nagad, cash), and status for accountability.
- **Delivery Method:** In-app receipt (screen display + downloadable PDF).
- **Design Justification:**
  - Serve the intended purpose: Provides proof of payment and reduces disputes.
  - Fit the user: Students can track costs for financial clarity.
  - Deliver the appropriate quantity: Shows one transaction per receipt; consolidated in logs.
  - Ensure output is where it is needed: Available immediately in payment history.
  - Provide output on time: Issued instantly after payment.
  - Choose the right output method: PDF/download option ensures permanence beyond the app.

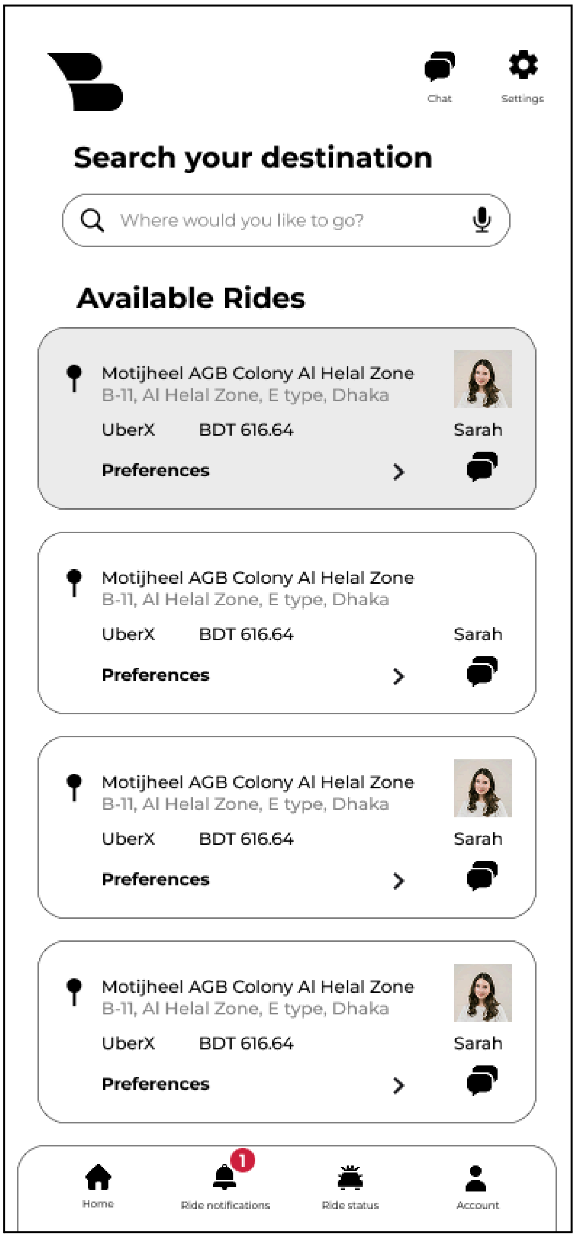
4. Ride History

- **Intended User:** Students
- **Primary Purpose:** Provide a record of all past rides (created, joined, cancelled) for personal tracking of transport usage.
- **Delivery Method:** In-app screen (history tab with searchable/filterable list).
- **Design Justification:**
  - Serve the intended purpose: Lets users reflect on their travel activity.
  - Fit the user: Organized tab view with filters (by date, role, or status).
  - Deliver the appropriate quantity: Past rides only, not cluttered with current rides.
  - Ensure output is where it is needed: Integrated in the personal profile section.
  - Provide output on time: Updated immediately after each ride completion.
  - Choose the right output method: In-app list view balances accessibility and clarity.

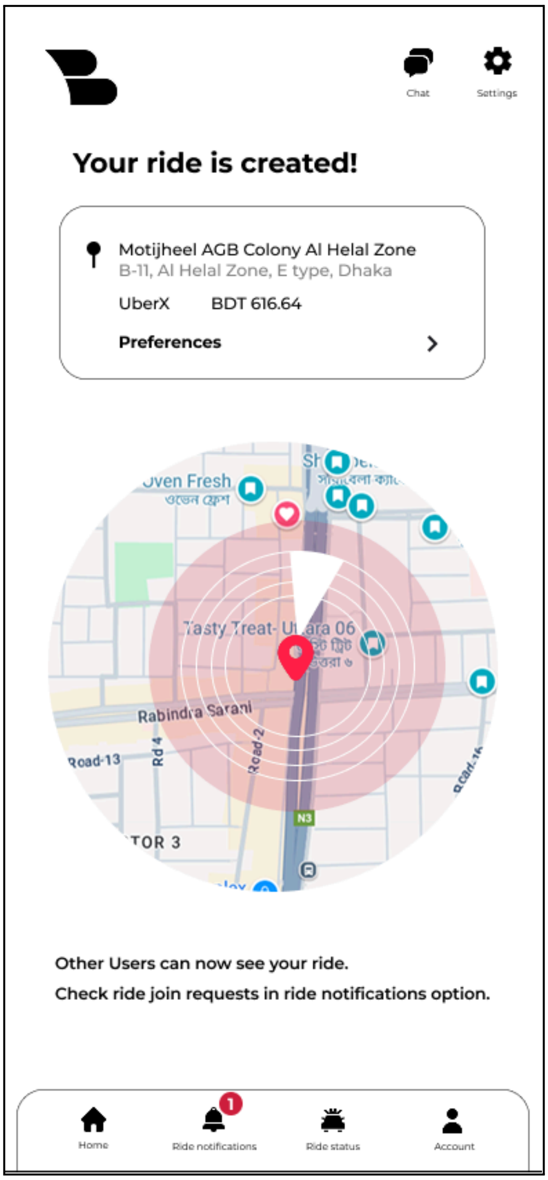
Core System Output Mockups

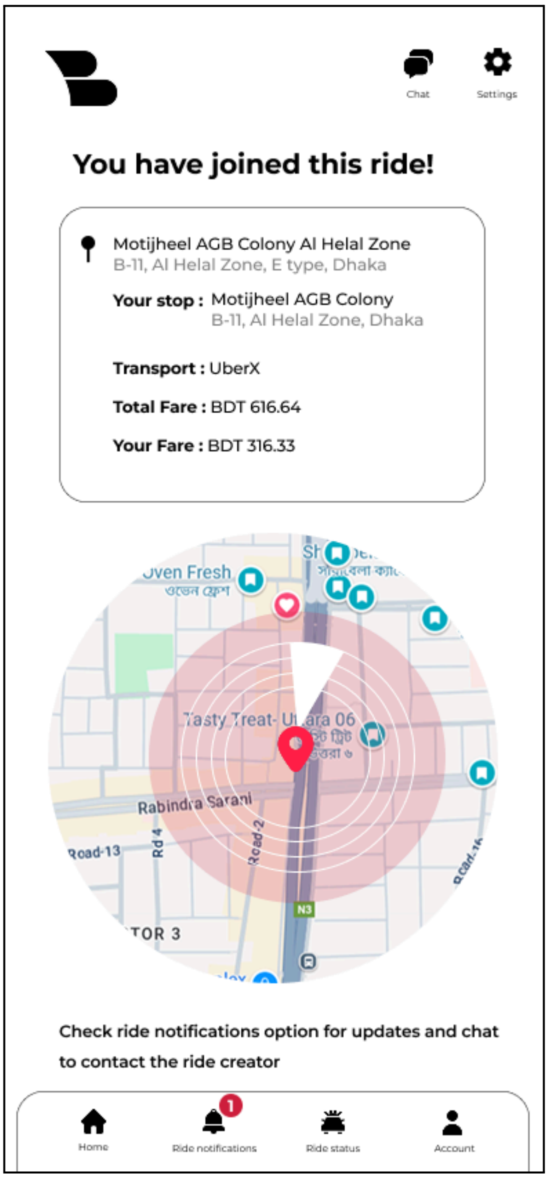
Core System Output 1: Ride Listings

- **Layout & Data Elements**: Displays available rides as cards populated from the Rides_Created table. Each card shows pickup location, drop-off, transport type (e.g., UberX), fare, and ride creator info. A "Preferences" drop-down allows users to view co-rider conditions.
- **Design Notes**: The card layout uses consistent spacing and alignment for readability. Icons (e.g., user profile) group visual information for quick scanning. Filtering and sorting are visually accessible, aligning with user focus on speed of decision-making.
- **User Fit & Bias Mitigation**: The design avoids clutter by summarizing key ride info up front. Bias from highlighting certain rides (e.g., by creator name) is reduced by presenting all cards in uniform style.



Core System Output 2: Ride Confirmation

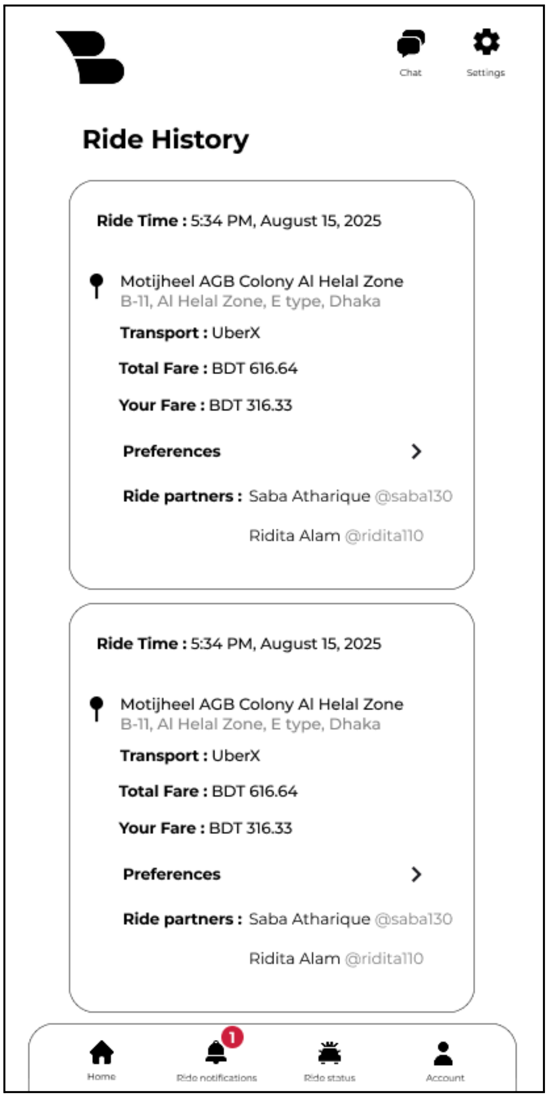




- **Layout & Data Elements**: Once a ride is created or joined, the app displays a confirmation page. Data comes from Rides_Created or Rides_Joined. Headings clearly state success ("Your ride is created" / "You have joined this ride"), followed by trip details (location, fare, stop, partner info). Status messages provide feedback.
- **Design Notes**: Ride details are grouped above the map visualization, ensuring users see confirmation before exploring route context. The alignment of text and map supports a natural top-to-bottom flow.

- **User Fit & Bias Mitigation**: Confirmation messages provide reassurance and reduce uncertainty. Maps mitigate misinterpretation of location text. Uniform phrasing prevents misleading emphasis on one outcome over another.

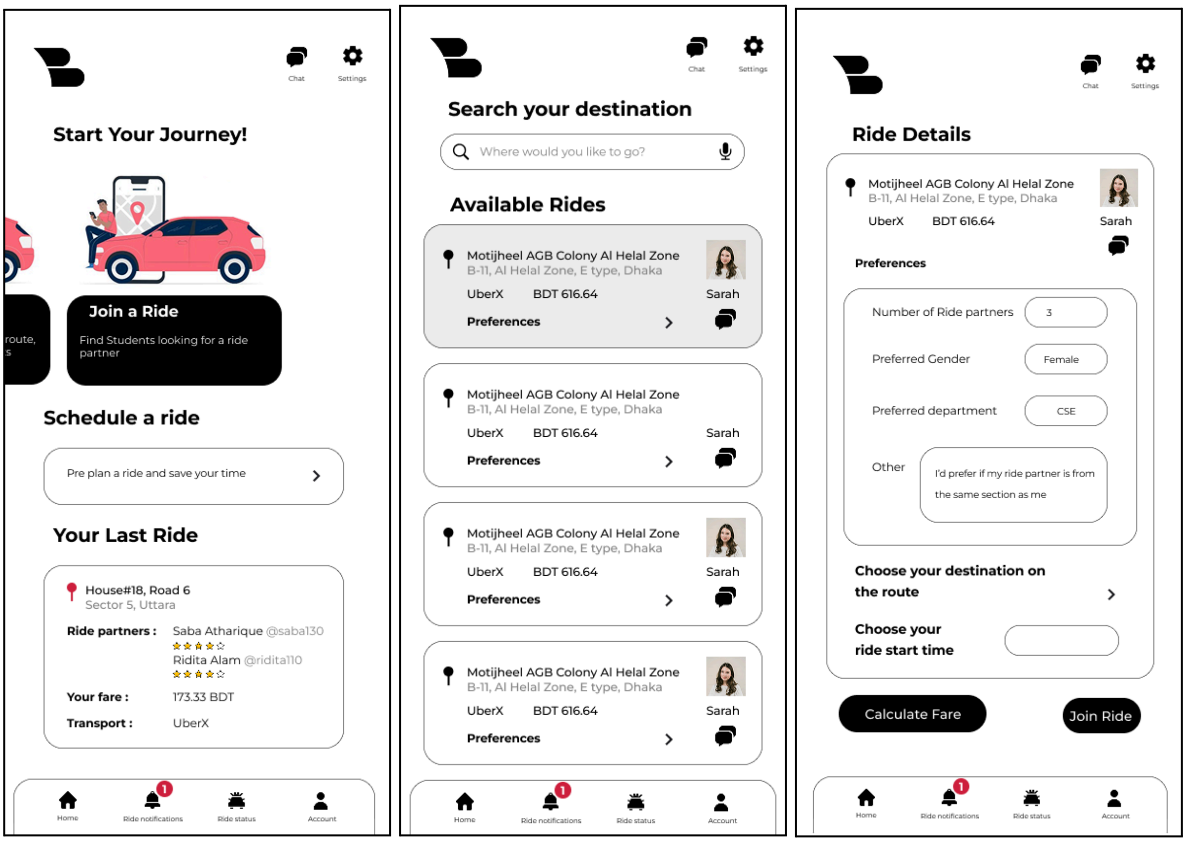
Core System Output 3: Ride History



- **Layout & Data Elements**: Past rides fetched from Rides_Created and Rides_Joined are displayed as cards in the History tab. Each entry includes ride time, fare breakdown (total and personal share), transport type, preferences, and partners.
- **Design Notes**: Cards are consistently structured, with ride time emphasized at the top. Preferences and ride partners are grouped at the bottom to provide context. Spacing between cards improves readability in long lists.
- **User Fit & Bias Mitigation**: The design supports easy personal tracking of transport use and financial accountability. Searchable and filterable fields prevent information overload. Neutral formatting ensures that no ride is visually highlighted over others.

Key Web/Mobile Interface Output

Selected Screen: Ride Listings & Ride Details (Mobile Interface)



Design Details:

- **Navigation Cues:** A bottom navigation bar with clear icons (Home, Notifications, Profile, Settings) allows users to move seamlessly between major functions. A consistent header with the logo and settings icon reinforces orientation.
- **Logical Grouping:**
  - **Available Rides:** Displayed as individual cards grouped under a heading. Each card shows location, fare, transport type, and the ride creator's profile picture, making quick scanning easier.
  - **Ride Details:** Preferences (partner count, gender, department) and route/ride time are grouped together in form fields, separated from action buttons ("Calculate Fare" and "Join Ride") for clarity.

- **Iconography & Visual Affordances:**
    - Profile pictures provide quick trust-building cues.
    - Dropdowns for “Preferences” and “Destination” afford further exploration without overwhelming the main view.
    - Buttons use bold, filled styles to signal primary actions.
- **Responsive Behavior:**
    - On smaller screens, cards stack vertically with scroll.
- **Aesthetic Style & Usability Goals:**
    - Clean, minimalistic layout with white background and black text ensures readability.
    - Accent icons and profile images break monotony and guide the eye.
    - Card-based design reduces cognitive load, letting students quickly compare rides.

Data Visualization Output

Scenario Definition (Use Case):
This visualization supports administrators and managers of the ride-sharing platform. Over a three-month observation period, they need to monitor how many rides are being created and how many students are joining rides on average each day. Identifying usage peaks and dips helps with resource allocation (e.g., server scaling) and strategic adjustments.

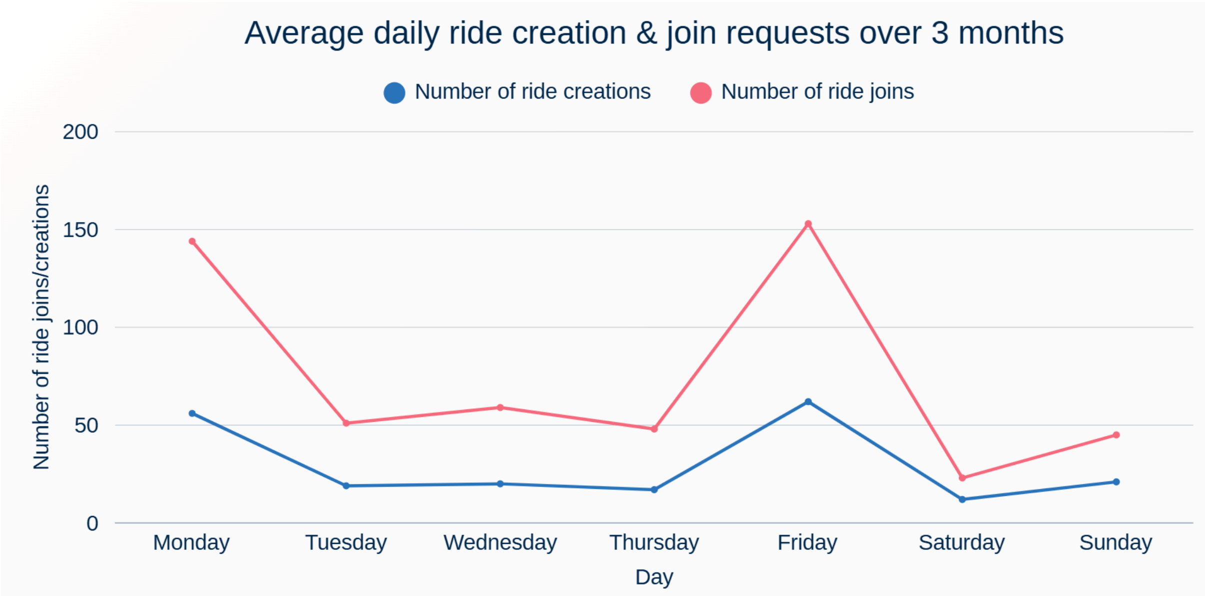


Chart Selection & Justification:

- **Chosen Chart:** Line Chart
- **Nature of Data:** Since the data represents time-series ride activity (daily averages across a week), line charts are ideal because they emphasize progression and directional trends over time.
- **Clarity:** Smooth connected lines with color-coded categories (creations vs joins) make it simple to compare two related but distinct flows in one view. Labeled axes and a descriptive title further enhance interpretability.
- **Efficiency:** A single glance reveals patterns such as high ride joins on Mondays and Fridays, dips on weekends, and steady lows midweek. This eliminates the need to manually scan through numeric reports or raw logs.
- **User Need Alignment:** Administrators can quickly identify demand surges (e.g., Monday/Friday peaks) and lows (e.g., Saturday), which supports timely decision-making for operational and promotional strategies.

Mockup Description:

- X-axis: Days of the week (Monday → Sunday).
- Y-axis: Number of ride creations/joins.
- Two colored lines: Blue (ride creations), Pink (ride joins).
- Title + subtitle explain the time scope and purpose of the chart.
- Visual grouping (legends, consistent spacing) improves readability and reduces cognitive load.